

The National University of Singapore
School of Computing

CS4226 – Internet Architecture

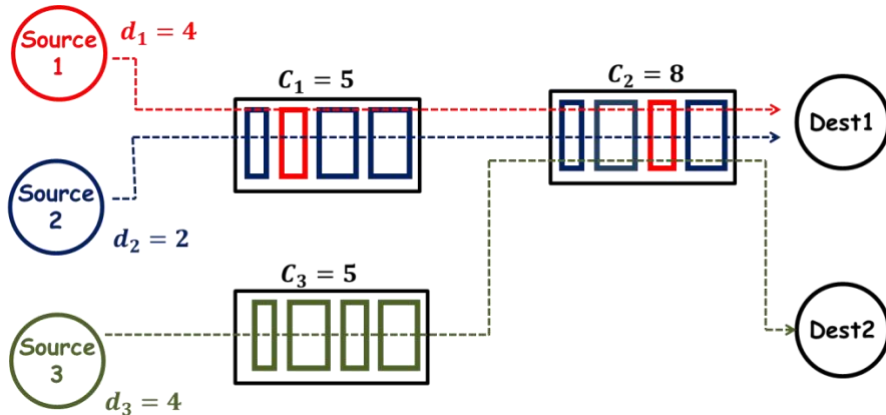
Tutorial 4

Week 6: September 14 - September 18, 2026

NOTE:

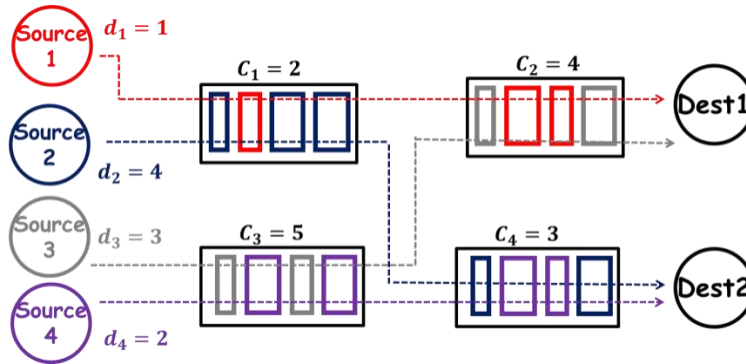
- Please work through the tutorial problems before class and bring your solutions with you.
- If any part of a question is unclear, you are encouraged to ask on Piazza.

Q1.



Consider the above topology with three flows and three links. The three flows have demands $d_1 = 4$, $d_2 = 2$ and $d_3 = 4$. The three links have capacities $C_1 = 5$, $C_2 = 8$ and $C_3 = 5$. Please identify the max-min fair allocation for the three flows and the bottleneck link(s).

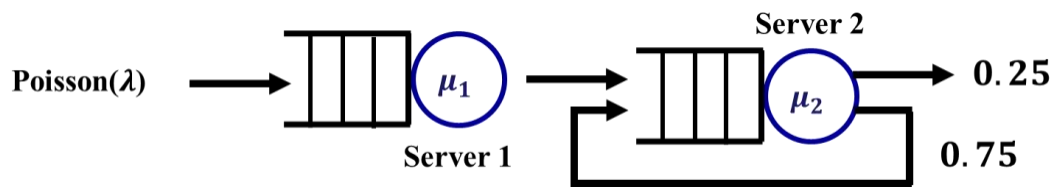
Q2.



Consider the above system with four flows and four links. The demand for the four flows are 1, 4, 3 and 2 (Mbps) and the capacity of the four links are 2, 4, 5 and 3 (Mbps), respectively.

- A. Calculate the max-min fair allocation $\mathbf{x} = (x_1, x_2, x_3, x_4)$ to the four flows.
- B. Calculate the weighted max-min fair allocation $\mathbf{x} = (x_1, x_2, x_3, x_4)$ to the four flows, when the weights are $\boldsymbol{\phi} = (\phi_1, \phi_2, \phi_3, \phi_4) = (2, 3, 4, 5)$.

Q3.



Consider the tandem queue structure where the arrival is Poisson with rate λ and the service times are exponential with rate μ_1 and μ_2 . After getting served by Server 2, each job has a probability of $1/4$ to leave the system and a probability of $3/4$ to join the queue of Server 2 again.

- How much processing capacities of μ_1 and μ_2 do we need to guarantee the stability of the system?
- Suppose we have a total capacity of $\mu = \mu_1 + \mu_2$, how can we optimally allocate the resource such that the average sojourn time of the jobs is minimized?